for granted, so long as it had merely the standing of a law disclosed by a lucky intuition, it could not be expected to possess more than a formal significance. For this reason, on the very day when I formulated this law, I began to devote myself to the task of investing it with a true physical meaning. This quest automatically led me to study the interrelation of entropy and probability—in other words, to pursue the line of thought inaugurated by Boltzmann. Since the entropy *S* is an additive magnitude but the probability *W* is a multiplicative one, I simply postulated that $S = k \cdot \log W$, where *k* is a universal constant; and I investigated whether the formula for *W,* which is obtained when *S* is replaced by its value corresponding to the above radiation law, could be interpreted as a measure of probability.

As a result,[1] I found that this was actually possible, and that in this connection *k* represents the so-called absolute gas constant, referred not to gram-molecules or mols, but to the real molecules. It is, understandably, often called *Boltzmann's constant.* However, this calls for the comment that Boltzmann never introduced this constant, nor, to the best of my knowledge, did he ever think of investigating its numerical value. For had he done so, he would have had to examine the matter of the number of the real atoms —a task, however, which he left to his colleague J. Loschmidt, while he, in his own calculations, always kept in sight the possibility that the kinetic theory of gases represents only a mechanical picture. He was therefore satisfied with stopping at the gram-atoms. The letter *k* has won acceptance only gradually. Even several years after its introduction, it was still customary to calculate with the Loschmidt number *L.*

Now as for the magnitude *W,* I found that in order to interpret it as a probability, it was necessary to introduce a universal constant, which I called *h.* Since it had the dimension of action (energy X time), I gave it the name, *elementary quantum of action.* Thus the nature of entropy as a measure of probability, in the sense indicated by Boltzmann, was established in the domain of radiation, too. This was made especially clear in a proposition, the validity of which my closest pupil, Max von Laue, convinced me in a number of conversations—namely, that the entropy of two coherent pencils of light is smaller than the sum of the entropies of the individual pencils of rays, quite consistently with the proposition that the probability of the happening of two mutually interdependent reactions is different from the product of the individual reactions.

While the significance of the quantum of action for the interrelation between entropy and probability was thus conclusively established, the part played by this new constant in the uniformly regular occurrence of physical processes still remained an open question. I therefore, tried immediately to weld the elementary quantum of action *h* somehow into the framework of the classical theory. But in the face of all such attempts, this constant showed